Igor Ignashin

Email: ignashin.in@phystech.edu
GitHub: github.com/ThunderstormXX

Education

- B.S. in Applied Mathematics and Physics, Moscow Institute of Physics and Technology (MIPT), Faculty of Applied Mathematics and Informatics.
- M.S. candidate, MIPT, Faculty of Applied Mathematics and Informatics, Department of Intelligent Data Analysis.

School Olympiad Achievements

- Winner, Phystech Mathematics Olympiad.
- Winner, Step into the Future Olympiads in Mathematics and Physics.
- Winner, KFU Interregional Mathematics Olympiad.
- Winner, municipal stage of the All-Russian School Olympiad in Mathematics.

Technical Skills

- **Languages:** Python, C++, SQL.
- **ML/Optimization:** PyTorch, JAX, RL, LLMs, convex and nonconvex optimization, game theory.
- **Mathematics:** stochastic processes, optimization, minimax problems, probability theory.
- **Data/Infra:** YQL, DataLens, Nirvana, Linux, Git.

Work Experience

Visiting Research Student, MBZUAI (under Eduard Gorbunov) Jan 2026 - Mar 2026

- Conducted theoretical research on LMO-based optimization methods and acceleration schemes, including analyses of the Muon optimizer and Lookahead-type techniques.

Laboratory of Mathematical Optimization Methods, MIPT / ISP RAS / BRAIn Lab 2024-Present

- Working on a Huawei project on neural network-based schedulers, with a focus on reducing the performance gap between classical and learned approaches.
- Proved local convergence of the ALSO method for nonconvex and strongly concave minimax problems in *Aligning Distributionally Robust Optimization with Practical Deep Learning Needs*.
- Leading student research projects at YSDA and MIPT AI360 on LLM inference acceleration, distillation, multi-agent RL, game-theoretic LLM training, SGD dynamics, and applied optimization, with an emphasis on theory, experiments, and publication-quality research.
- Collaborating with Andrei Leonidov's group on SGD analysis and multi-agent reinforcement learning, including SGD noise analysis, theoretical frameworks for nonconvex optimization dynamics, and joint research on multi-agent LLM systems in game-theoretic scenarios and social dilemmas.
- Researched equilibrium traffic assignment problems and Frank-Wolfe algorithm modifications, resulting in two journal publications and two conference presentations. Currently developing stochastic extensions of the LMO oracle and exploring applications to game-theoretic formulations.

Yandex, Data Analyst Intern June 2022 - September 2022

- Automated a daily pipeline for ticket distribution and closure analytics for B2B contractors, improving transparency and monitoring efficiency.
- Built data processing workflows in YQL and designed dashboards in DataLens for operational insights.
- Worked with Yandex internal tools, including Nirvana, for orchestrating data workflows.

Selected Publications

Google Scholar profile

- Aligning Distributionally Robust Optimization with Practical Deep Learning Needs. Accepted to a NeurIPS 2025 workshop.
- NeurIPS 2026 submission on stochastic dynamics of SGD, proposing a new perspective beyond Brownian-motion assumptions: *Why SGD is not Brownian Motion: A New Perspective on Stochastic Dynamics*.
- Publication in *Computer Research and Modeling*: Frank-Wolfe Modifications for Equilibrium Traffic Assignment.
- Preprint: Stochastic Origin Frank-Wolfe for Traffic Assignment. Presented at TFN-2025. Accepted to the Journal of Mathematical Sciences, Series B.
- Preprint: Modeling Skier Flows via Wardrop Equilibrium in Closed Capacitated Networks. Presented at TFN-2025. Accepted to the Journal of Mathematical Sciences, Series B.
- *Conjugate Frank-Wolfe in Machine Learning.* Presented at OPTIMA and accepted to CCIS.

Summer Schools and Hackathons

- AIRI Institute Summer School on Artificial Intelligence 2025.
- Summer School "Control, Information and Optimization" in memory of B. T. Polyak 2025.
- Summer School and Hackathon in Financial Mathematics, Quantathon 2024.
- 5th and 6th International Summer Schools on Financial Mathematics (2024-2025).